

TCGA Pan-Cancer Derived Prognostic Signature Report

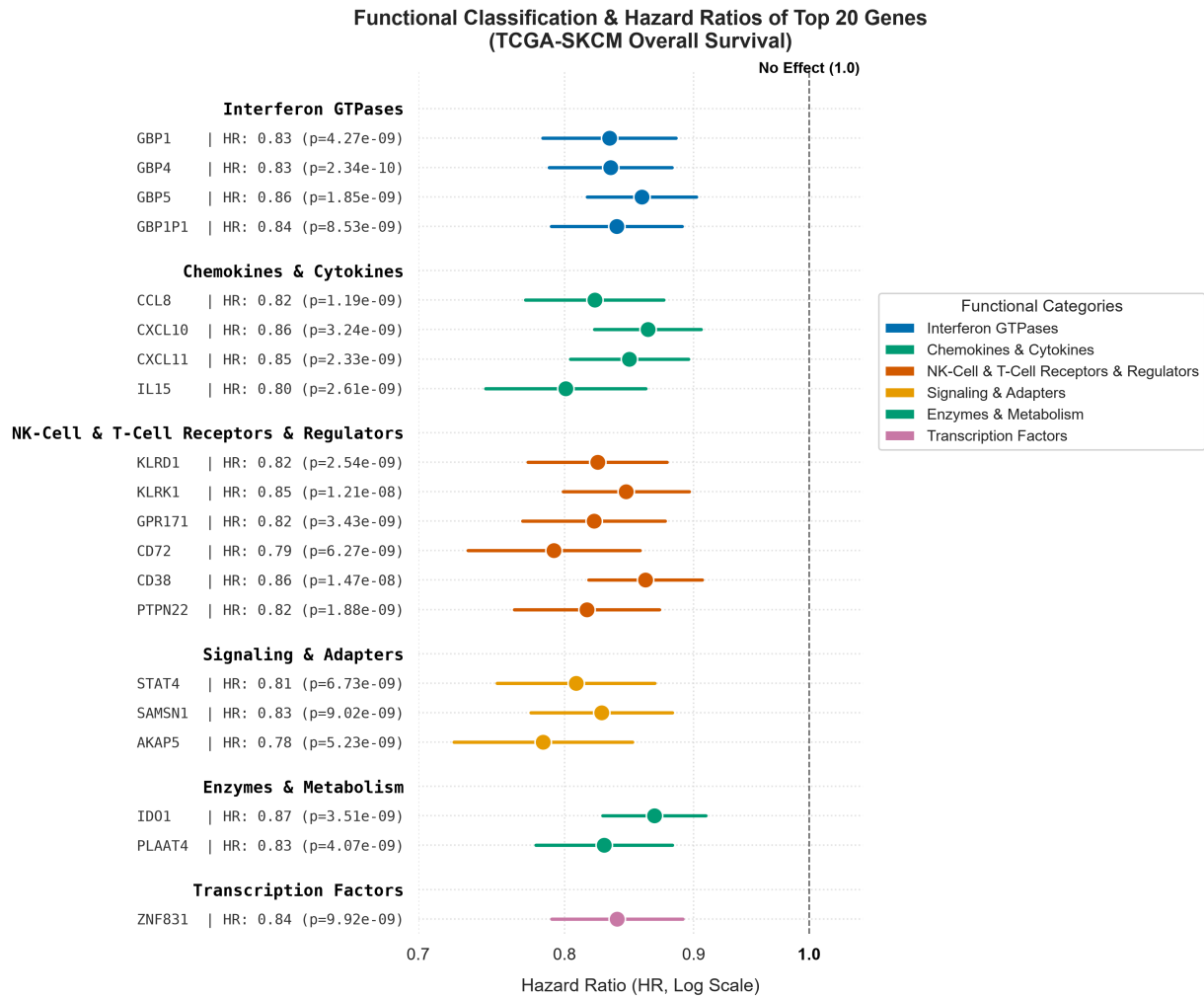
We performed transcriptomic feature selection on the **TCGA-SKCM** cohort ($N = 428$ aligned samples with survival data) to build a custom overall survival signature, and subsequently validated it on three independent clinical trial cohorts.

1. Top 20 Prognostic Genes in TCGA-SKCM

The 20 genes most significantly associated with overall survival in univariate Cox regression are visualised below. A positive Beta indicates a **risk-associated gene** (higher expression = worse survival), while a negative Beta indicates a **protective gene** (higher expression = better survival).

Hazard Ratio Forest Plot (Top 20 Genes)

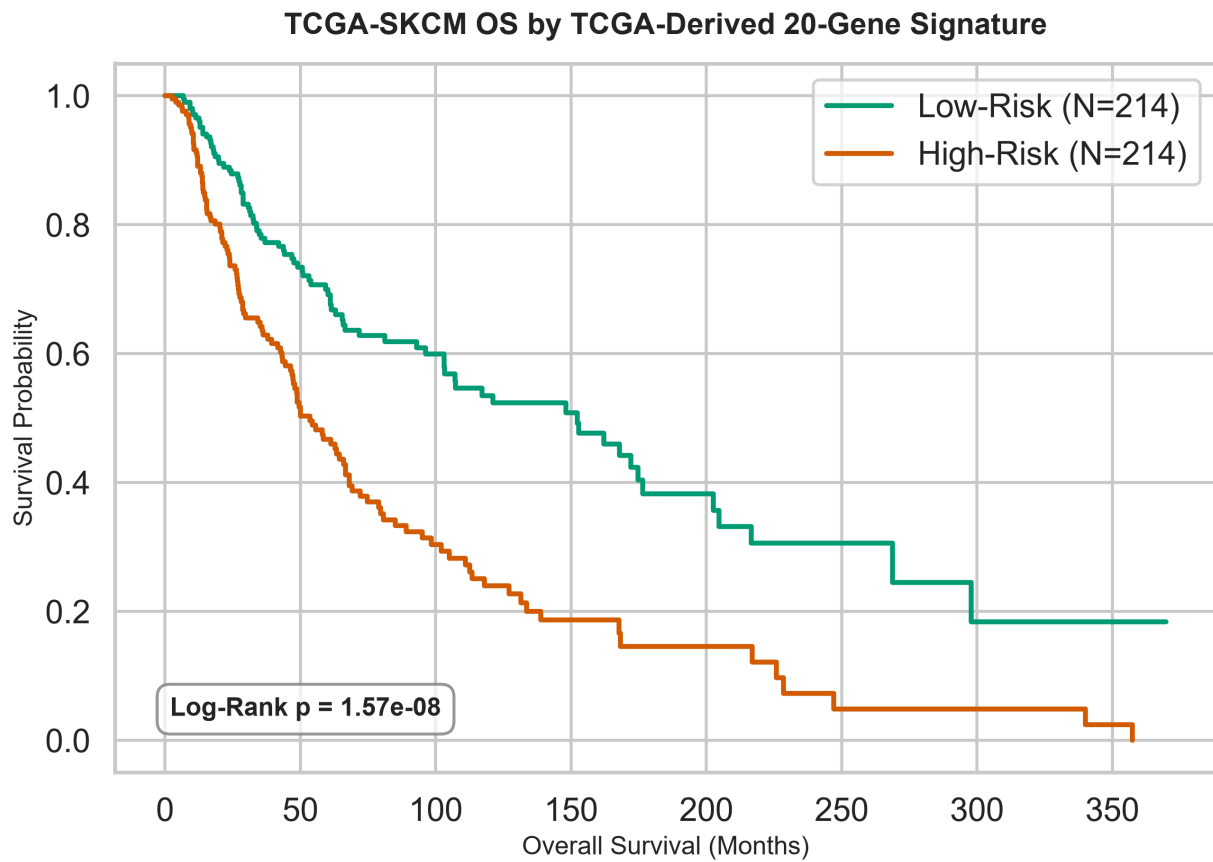
The forest plot below visualises the Hazard Ratios (HR) and their 95% confidence intervals for the top 20 most significant prognostic transcripts. Protective genes ($HR < 1.0$) are shown in blue, and risk-associated genes ($HR > 1.0$) are shown in red:



2. Kaplan-Meier Survival Curve on TCGA

We partitioned TCGA-SKCM patients into High-Risk and Low-Risk groups using the median value of the signature score. The log-rank test indicates an extremely significant separation in survival curves:

- **Log-Rank p-value: 1.57e-08**



3. Validation on Immunotherapy Clinical Trial Cohorts

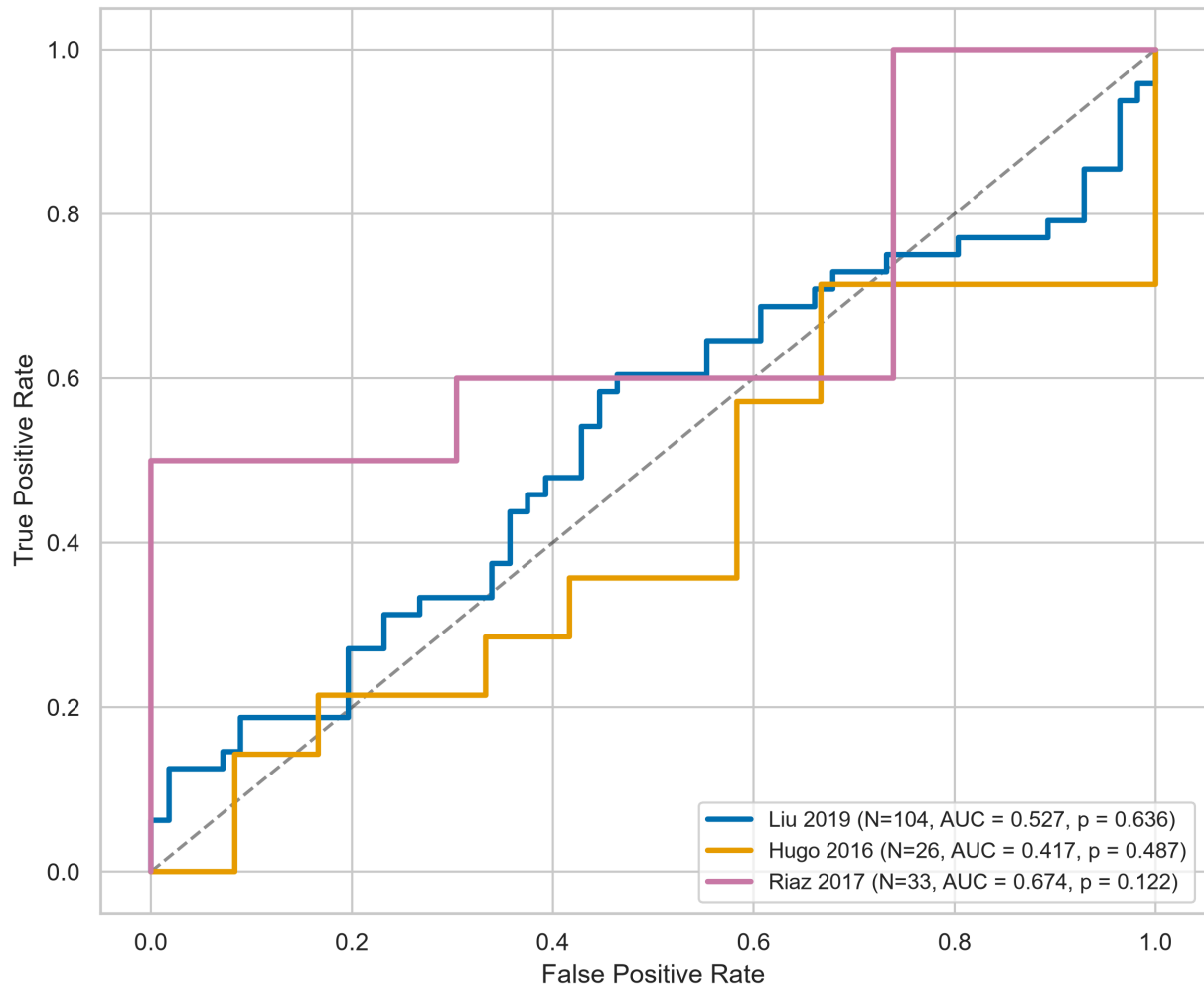
We evaluated the custom 20-gene prognostic signature on three cohorts receiving anti-PD-1 or combination immunotherapies to see if the overall survival signature translates into predicting immunotherapy response.

Cohort	N	Aligned Signature Genes	Response ROC AUC	Mann-Whitney U p-value	Mean Risk (Responders)	Mean Risk (Non-Responders)
Liu 2019	104	20/20	0.527	6.36e-01	-18.067	-18.224
Hugo 2016	26	20/20	0.417	4.87e-01	-18.842	-19.102
Riaz 2017	33	20/20	0.674	1.22e-01	-19.225	-18.652

Validation Visualisations

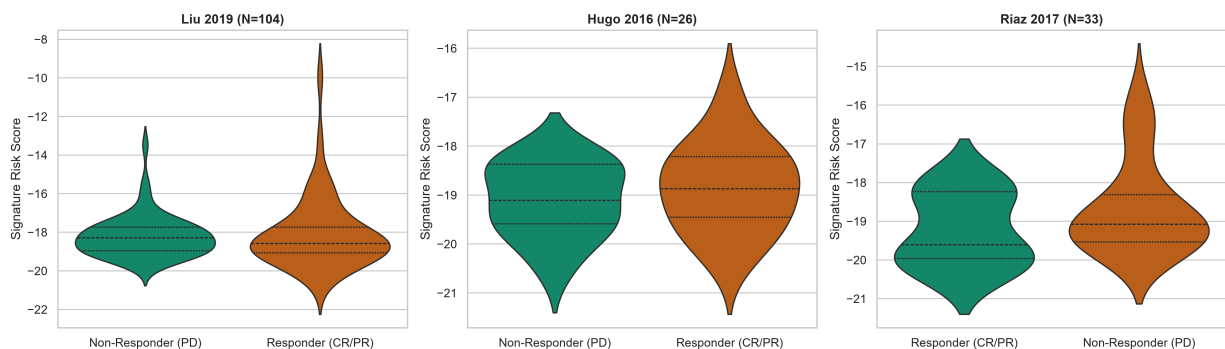
ROC Curves Predicting Response

Trial Validation: Predicting Immunotherapy Response



Signature Risk Score Stratified by Responders vs. Non-Responders

Signature Risk Score Stratified by Immunotherapy Response



4. Part 3: Synthesis & Pipeline Recommendation

What, Why & Key Questions

What: We evaluated whether a custom 20-gene overall survival score built from TCGA-SKCM data should be added to our final prediction model alongside our 6 established immune signatures.

Why: We must ensure every feature added to our machine learning model provides unique, genuine predictive value rather than repeating information or adding random noise.

Key Finding: We decided **not** to include the TCGA survival score in our final model because it measures the exact same underlying immune signal as our existing signatures and fails to predict treatment response in real-world patient trials.

1. Why Purely Data-Driven Gene Selection Fails:

Selecting genes purely by statistical algorithms (`SelectKBest`) picks up trial-specific noise (like nerve or stomach genes) that do not generalise to new patients. In contrast, using established, biologically curated immune signatures (like IFN- γ and TIS) condenses over 20,000 raw genes into 6 meaningful, reliable biological scores ($D = 6$).

2. Why We Do Not Need the TCGA Survival Score:

- **Biological Overlap:** Even though the 20 genes in the TCGA survival score are different names from the genes in our 6 immune signatures (0/20 literal gene overlap), they perform the exact same job. They are all helper genes turned on by interferon to activate the immune system.
- **Strong Correlation:** Measuring the TCGA score against our immune signatures gives near-identical results across all patient datasets ($r_s = -0.83$ to -0.94 with TIS and IFN- γ). Adding it would simply count the same biological signal twice.
- **Poor Response Prediction:** When tested on actual immunotherapy patient trials, the TCGA survival score could not reliably tell responders from non-responders in any cohort (**Liu 2019**: AUC = 0.527, $p = 0.636$; **Hugo 2016**: AUC = 0.417, $p = 0.487$; **Riaz 2017 Pre-treatment**: AUC = 0.674, $p = 0.122$). General patient survival in TCGA does not equal immunotherapy success.

🎯 Final Pipeline Architecture Decision

Because the TCGA survival score is biologically redundant and fails to predict treatment response out-of-cohort, we **exclude** it from our final machine learning pipeline. We use only the **6 curated functional immune signatures** (IFN- γ , TIS, CYT, CD8 T-cell, IMPRES, PD-L1) for transcriptomic features ($D = 6$). Combined with our clinical and genomic variables, this builds our final 14-feature multimodal matrix ($D = 14$).